| MEASUREMENT |
| SENSITIVE |

MIL-B-28635D
31 July 1990
SUPERSEDING
MIL-B-28635C(YD)
22 September 1985

MILITARY SPECIFICATION

BOOKSHELF FOR BUNK BED

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements for component materials and their assembly into bookshelves.

1.2 Classification The bookshelves shall be of the following types, as specified (see 6.2).

Type I - Bookshelf for round tube bunk bed (see figures 1 and 2).
Type II - Bookshelf for square tube bunk bed (see figure 3).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7105

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2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1 1 Specifications and standards The following specifications and standards form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

MMM-A-130 - Adhesive, Contact.
MMM-A-180 - Adhesive, Polyvinyl Acetate Emulsion
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-15228 - Bed, Bunk, Steel, Round Tube, Single and Double-Deck (Convertible).
MIL-B-28644 - Bed, Bunk, Steel, Square Tube, with End Panels, Single and Double-Deck.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-2073-1 - DOD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI A208.1 - Mat-Formed Wood Particleboard.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

ASTM

ASTM A 366 - Steel Sheet, Carbon, Cold-Rolled, Commercial Quality, Specification for.
ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA LD 3 - High-Pressure Decorative Laminates.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a bookshelf of the type specified for first article inspection and approval (see 4 2.1 and 6.4).

3.2 Materials. Materials shall be as specified herein and in applicable specifications and standards, and other referenced documents. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification. Materials shall be free of defects which adversely affect performance or serviceability of the finished product.

3.2.1 Particleboard. Particleboard shall be 3/4 inch thick, and in accordance with ANSI A208.1, type 1, grade 1-M-3.

3.2.2 Decorative laminated plastic sheets. Decorative laminated plastic sheets shall be a minimum of 0.032-inch thick and in accordance with NEMA LD 3. Pattern, color and finish shall be as specified (see 6.2).

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3.2.3 Plastic edge molding. Plastic edge molding shall be tee-style, extruded of ethyl cellulose, polypropylene, or cellulose acetate butyrate The tongue of the tee shall be formed with barbs to hold securely in the slot of panel edge. Face shall be convex and smooth, without grooves or other detailing. Edge molding shall match the wood grain of the decorative laminated plastic sheet or be a solid color which blends with the woodgrain color, as specified (see 6 2). When solid color laminated plastic sheets are used, the edge molding shall be the same color or a lighter shade of the same color.

3.2.4 Bracket fasteners Hardware for securing the two brackets to the type I bookshelf shall consist of cap screws with button heads and hexagon socket openings, capped tee-nuts, and washers. The capped tee-nuts shall be pronged to prevent turning during assembly. Hardware dimensions shall be as shown on figure 2 and finish shall be cadmium plating for corrosion resistance.

3 2.5 Particleboard mechanical fasteners The particleboard back and bottom of the type I bookshelf shall be secured to the sides with helically wound coils of carbon steel wire The wire insert shall be of double coarse thread type with matching screw

3 2 6 Sheet steel Sheet steel for the type I brackets, and the shelf and channel of the type II bookshelf shall conform to ASTM A 366. Sheets of 16-gauge and lighter shall have a stretcher-leveled standard of flatness.

3.2.7 Adhesive. Adhesive for bonding the decorative laminated plastic sheet to particleboard shall conform to MMM-A-130 or MMM-A-180.

3.3 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts.

3.4 Design and construction The bookshelves shall be constructed as specified herein and as shown on figures 1 through 3 for the type specified Allowable tolerances shall be as shown on figures 1 through 3 and as specified herein Materials shall be in accordance with 3.2 and as otherwise specified herein.

3.4.1 Brackets.

3.4.1.1 Type I brackets. Each type I bookshelf shall have two brackets of 10-gauge steel. The brackets shall be formed and punched as shown on figures 1 and 2. Sharp edges shall be eased and burrs shall be removed. When finished as specified in 3.6, the brackets shall fit snugly over the headboard corner posts.

3 4 1 1.1 Type I bracket assembly. The plastic surrounding the opening for the capped tee-nuts shall be punched as necessary at the proper locations for insertion of the three prongs of the capped tee-nuts. The two brackets and shelf shall be assembled with a washer being placed between the cap screw head and bracket, and two washers placed between the bracket and shelf side as shown on figure 2. The two washers are used as spacers and may be removed as

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necessary for adjustment if there is any variation in distance between the corner posts. When the type I bookshelf is completely assembled it shall be capable of being installed or removed without the use of tools.

3.4.1.2 Type II brackets. Each type II bookshelf shall have a channel bracket of 10-gauge steel spot welded to the underside of the shelf at the front. Sharp edges shall be eased and burrs shall be removed. When finished as specified in 3 6, the bracket shall fit snugly over the uppermost crossmember of the headboard.

3.4 2 Panels.

3.4.2.1 Type I panels. The type I bookshelf shall be constructed of particleboard panels. The panels consisting of the bottom, back, and two sides shall be covered on both sides with decorative laminated plastic sheets. The exposed edges shall be covered with plastic edge moldings in accordance with 3.5 Laminated plastic sheets shall be bonded to the particleboard core as specified in 3.4.3. The finished laminated plastic surfaces shall not contain cracks, scratches, dents, or chipped edges. Slots in the shelf bottom for the exit of electrical cords and for use in cleaning shall be of the size and location shown on figure 1. Exposed particleboard edges created by the slots shall be stained to closely match the adjoining plastic surfacing. The back, bottom and side of each end of the bookshelf shall be secured with four wire inserts and matching screws at the locations shown on figure 2. The back shall be fastened to the bottom with three wire inserts and matching screws; one centered between the ends, and one near each end of the bottom between the opening and the bookshelf side. The screws which fasten the sides and back to the shelf bottom shall be installed prior to bonding the laminated plastic sheets to the particleboard to provide a smooth surface without holes in the plastic surface. Joining of exterior surfaces of the panels shall fit flush.

3.4.2 2 Type II panels. The type II bookshelf shall be constructed of particleboard back and end panels, and a bottom panel of 16-gauge sheet steel. The particleboard panels shall be covered on both sides with decorative laminated plastic sheets, and the exposed edges covered with plastic edge moldings in accordance with 3.5. Laminated plastic sheets shall be bonded to the particleboard core as specified in 3.4.3. The finished laminated plastic surfaces shall not contain cracks, scratches, dents, or chipped edges. Slots in the bottom panel for the exit of electrical cords and for use in cleaning shall be of the size and location shown on figure 3. The bottom panel shall be finished as specified in 3.6. Assembly of the type II bookshelf shall be as shown on figure 3. When the type II bookshelf is completely assembled it shall be capable of being installed or removed without the use of tools.

3 4.3 Bonding. Laminated plastic sheets shall be bonded to the particleboard core with a solvent based contact adhesive as specified in 3.2.7. The bonding procedure shall be in accordance with NEMA LD 3.

3.5 Installation of the plastic edge molding. The molding shall cover particleboard panel edges with a tolerance of plus 0-inch and minus 1/64-inch on each side. Molding for each exposed panel edge shall be one continuous piece, without splicing; except that one butt joint centered at the bottom is permitted for each side panel molding. Both ends of each length of molding shall be

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secured with brads driven through the molding tongue. Brads for securing the bottom and the back panel edge moldings shall be driven from the least exposed surfaces· that is, from the underside of the bottom and the outside of the back respectively. Brads for securing side panel molding shall be driven from the exterior sides. Brads shall be set, and holes filled completely with a nonshrinking, pigmented filler material to closely match the plastic surfacing The molding shall fit tightly and securely, without gaps or crevices.

3.6 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to ensure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 1.1 mils over the entire surface if an epoxy powder is used The painting shall consist of at least one coat of primer and one finish coat if a baking enamel is used. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. If baking enamel is used, the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.7 Identification marking. Identification shall be permanently and legibly marked directly on the underside of the shelf or on a corrosion-resisting metal plate securely attached to the underside of the shelf at the source of manufacturer. Identification shall include the manufacturer's model number, name and trademark to be readily identifiable to the manufacturer.

3 8 Workmanship.

3 8.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

3.8.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.8 3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot welds shall be a minimum of 0.016-inch diameter when measured at the fused nugget or 0 8 times the square root of the material thickness, whichever is greater. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4 2.1 First article inspection The first article inspection shall be performed on one bookshelf when a first article is required (see 3.1 and 6 2) This inspection shall include the examination of 4 4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4 and the packaging inspection of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105 The unit of product shall be one bookshelf. All bookshelves offered for delivery at one time shall be considered a lot for the purpose of inspection (see 6.3)

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4.4 Examination Each bookshelf selected shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements Noncompliance with any specified requirement shall constitute one defect

4.5 Packaging inspection. The preservation, packing, palletization and marking shall be inspected to verify conformance with section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1 1 Level A Each bookshelf shall be preserved method III in accordance with MIL-P-116 in a close-fitting fiberboard box conforming to PPP-B-636, class weather-resistant Bookshelves shall be shipped completely assembled. The contents shall be cushioned, blocked and braced to prevent movement. The fiberboard box shall be closed in accordance with method V as specified in the appendix to the box specification

5.1.2 Commercial. Material shall be preserved in accordance with ASTM D 3951. Bookshelves shall be shipped completely assembled.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5 2.1 Levels A and B Packing shall be in accordance with MIL-STD-2073-1 for the applicable level specified. Containers shall be selected from table VII, appendix C, for the appropriate level Only closed containers shall be used.

5.2 2 Commercial Material shall be packed in accordance with ASTM D 3951.

5.3 Palletization. Material shall be palletized in accordance with MIL-STD-147 when the following criteria are met:

a. Load to consist of four or more unskidded containers; and,
b. Load shall utilize a minimum of 80 percent of the pallet base.

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory)

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6 1 Intended use. The bookshelves are for use with military standard bunk beds for enlisted personnel conforming to Military Specification MIL-B-15228 or MIL-B-28644. One bookshelf is required per bed The type I bookshelf slips directly over the headboard corner posts and rests on the upper cross tube. The steel channel bracket of the type II bookshelf slips directly over the uppermost crossmember of the headboard and the ends rest against the headboard corner posts.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Whether type I or type II is required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 1 and 2.2).
d When a first article is required (see 3 1, 4 2.1 and 6 5).
e. Pattern, color and finish of decorative laminated plastic sheets required (see 3 2 2)
f. Whether the edge molding is to be wood grain color or a solid color (see 3.2 3).
g. Color of the finish coat (see 3 6).
h. Level of preservation and level of packing required (see 5.1 and 5.2).

6.3 Sampling procedures. Inspection level should be level II and the Acceptable Quality Level should be 2.5 percent defective If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots should be reinspected using tightened inspection. If the rejected lot was screened, reinspection may be limited to the defect causing rejection. If the lot was reprocessed, reinspection may be performed for all defects. Rejected lots should be separated from new lots, and should be clearly identified as reinspected lots (see 4 3)

6 4 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of the DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.5 First article. When a first article inspection is required, the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4 2 1 The first article should consist of one complete bookshelf. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.6 Subject term (key word) listing.

Brackets
Panel
Steel
Tube

6 7 Changes from previous issue Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - YD
Air Force - 99

Review Activity.
Air Force - 84

User Activity·
Army - GL

Preparing Activity:
Navy - YD

(Project 7105-0182)

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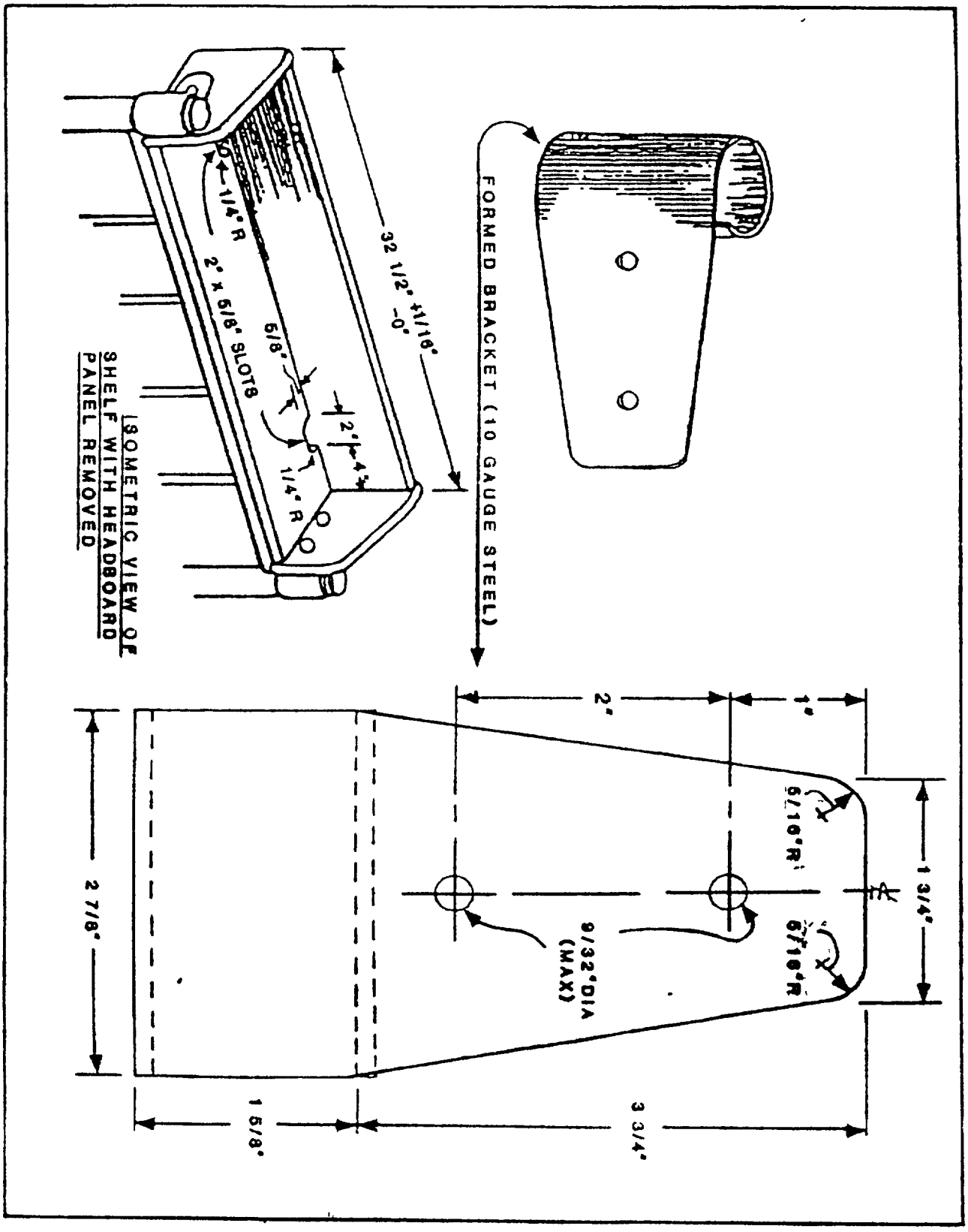


FIGURE 1. Type I bookshelf - details.

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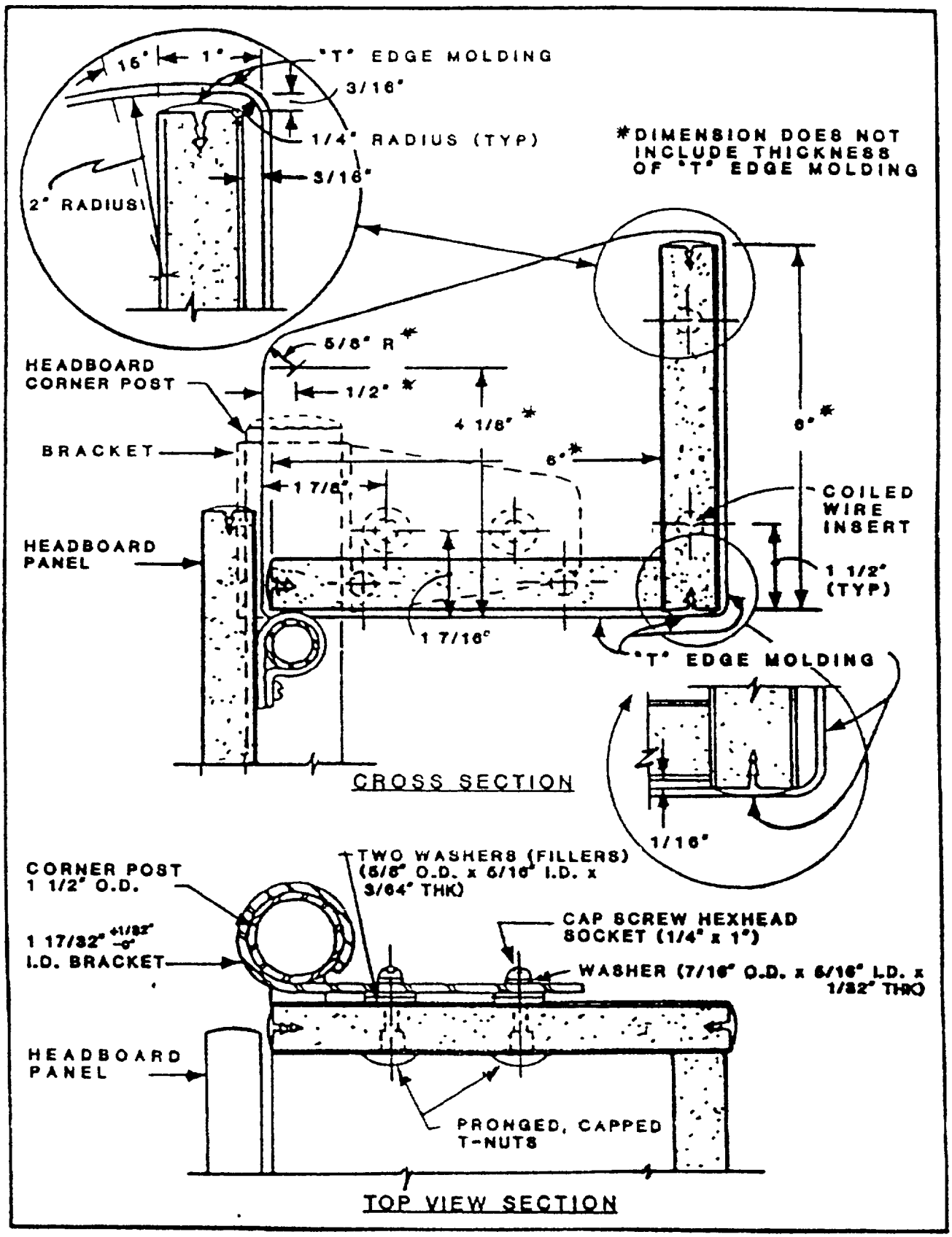
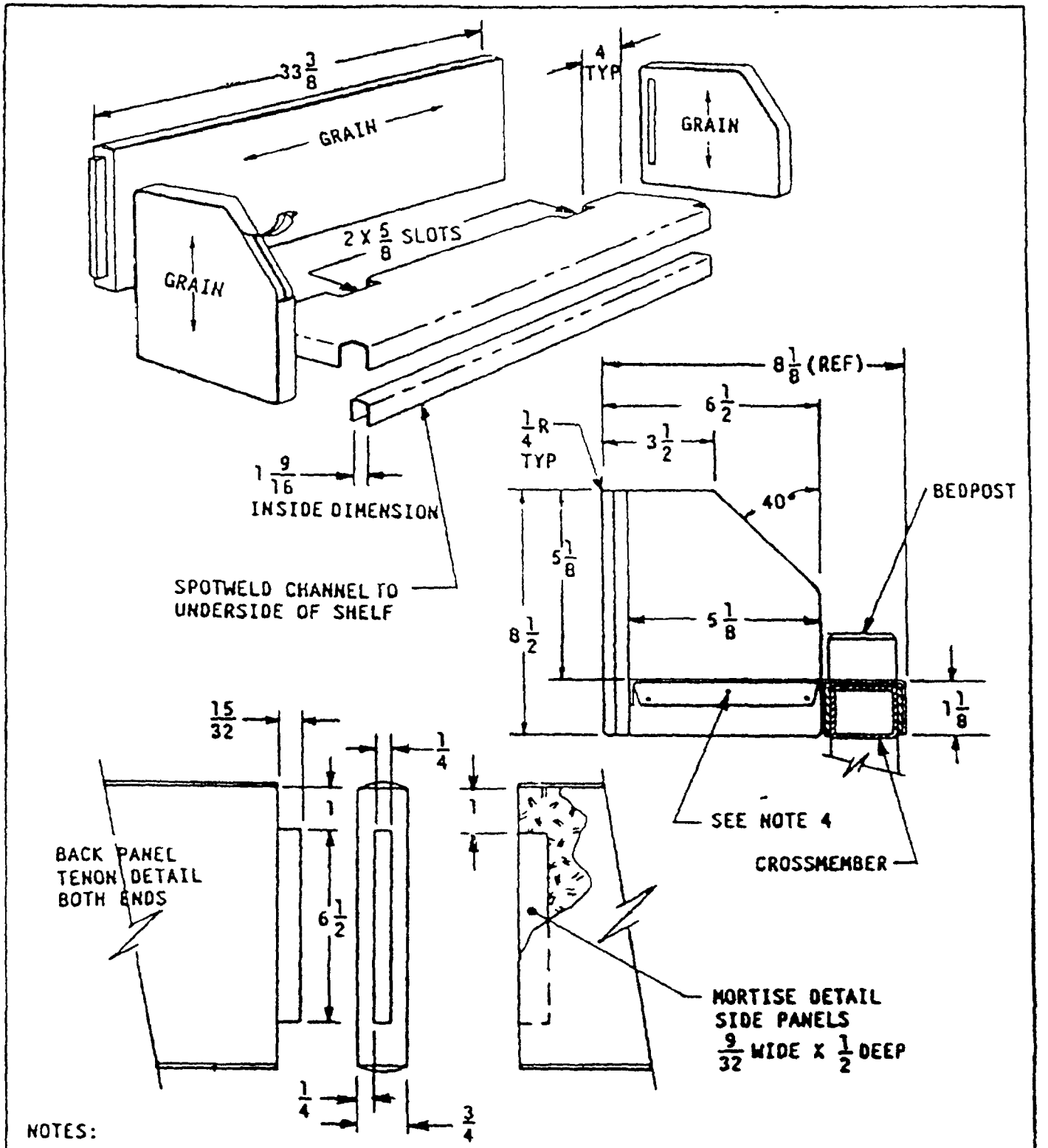


FIGURE 2. Type I bookshelf - sections.

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NOTES:

1. DIMENSIONS ARE INCHES. DIMENSIONS DO NOT INCLUDE "T" MOLDING.
2. APPLY PLASTIC LAMINATE & "T" MOLDING TO ENDS AND BACK AS SHOWN (BOTH SIDES).
3. ASSEMBLE & GLUE ENDS AND BACK PANEL.
4. LOCATE SHELF ASSEMBLY AS SHOWN. DRILL AND FASTEN WITH NO.10 PAN HEAD SCREWS.

FIGURE 3. Type II bookshelf.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-28635D	90/07/31

3 DOCUMENT TITLE
Bookshelf for Bunk Bed

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME L. LEWIS, Project Manager EQUIPMENT CRITERIA DEVELOPMENT DIV.	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (805) 982-5801 ext 456 551-5801 ext 456
c. ADDRESS (Include Zip Code) Commanding Officer Naval Construction Battalion Center Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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